

Mode-1 Update in 3D ACE–Tucker Alternating Least Squares

Model Setup

For each configuration $s = 1, \dots, S$, we define ACE features

$$a_x^{(s)} \in \mathbb{R}^{d_x+1}, \quad a_y^{(s)} \in \mathbb{R}^{d_y+1}, \quad a_z^{(s)} \in \mathbb{R}^{d_z+1}.$$

We approximate the energy using a Tucker decomposition:

$$E^{(s)} \approx \sum_{p=1}^{R_x} \sum_{q=1}^{R_y} \sum_{r=1}^{R_z} X_p^{(s)} Y_q^{(s)} Z_r^{(s)} \mathcal{G}_{pqr},$$

where

$$X^{(s)} = A^\top a_x^{(s)}, \quad Y^{(s)} = B^\top a_y^{(s)}, \quad Z^{(s)} = D^\top a_z^{(s)}.$$

Here,

$$A \in \mathbb{R}^{(d_x+1) \times R_x}, \quad B \in \mathbb{R}^{(d_y+1) \times R_y}, \quad D \in \mathbb{R}^{(d_z+1) \times R_z}.$$

Mode-1 Update (Updating A)

Fix B and D . Define

$$Y^{(s)} = B^\top a_y^{(s)}, \quad Z^{(s)} = D^\top a_z^{(s)}.$$

We rewrite the model as linear in $a_x^{(s)}$:

$$E^{(s)} \approx \sum_{k=0}^{d_x} \sum_{q=1}^{R_y} \sum_{r=1}^{R_z} a_x^{(s)}(k) \tilde{\mathcal{G}}_{kqr} Y_q^{(s)} Z_r^{(s)}.$$

This is linear in the unknown tensor

$$\tilde{\mathcal{G}} \in \mathbb{R}^{(d_x+1) \times R_y \times R_z}.$$

Stacking over all samples gives a least-squares problem:

$$\min_{\tilde{\mathcal{G}}} \sum_{s=1}^S \left(E^{(s)} - \sum_{k,q,r} a_x^{(s)}(k) \tilde{\mathcal{G}}_{kqr} Y_q^{(s)} Z_r^{(s)} \right)^2.$$

Low-Rank Projection

Reshape $\tilde{\mathcal{G}}$ into a matrix:

$$\tilde{G}_{(1)} \in \mathbb{R}^{(d_x+1) \times (R_y R_z)}.$$

Compute its truncated SVD:

$$\tilde{G}_{(1)} = U \Sigma V^\top.$$

Take the leading R_x singular vectors:

$$A = U_{[:,1:R_x]}.$$

Metric Orthonormalization

Since ACE features are not orthonormal, we define the Gram matrix

$$G_x = \sum_{s=1}^S a_x^{(s)} a_x^{(s)\top}.$$

We enforce

$$A^\top G_x A = I,$$

via Cholesky-based orthonormalization.

Core Update

The updated core tensor is obtained from the projection:

$$\mathcal{G}_{(1)} = A^\top G_x \tilde{\mathcal{G}}_{(1)}.$$

Reshaping $\mathcal{G}_{(1)}$ gives the new core

$$\mathcal{G} \in \mathbb{R}^{R_x \times R_y \times R_z}.$$

Summary

The mode-1 ALS step consists of:

1. Solve a linear least-squares problem for $\tilde{\mathcal{G}}$.
2. Perform truncated SVD of its mode-1 unfolding.
3. Set A to leading singular vectors.
4. Enforce $A^\top G_x A = I$.
5. Update the core by projection.

This yields the optimal rank- R_x update of A given fixed B and D .